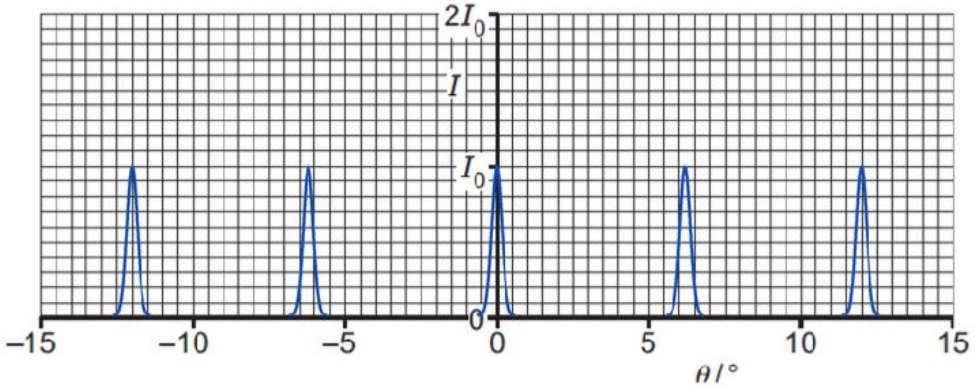
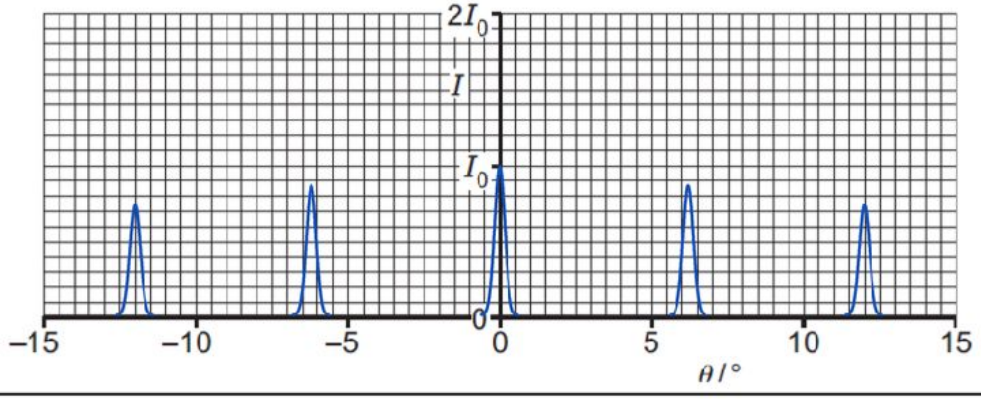
5(a)(i)	$d\sin\theta = n\lambda$ $\theta = \sin^{-1}[(540 \times 10^{-9})/(5.0 \times 10^{-6})]$ $= 6.2°$	[1]
5(a)(ii)	$d\sin\theta = n\lambda$ $\lambda = \sin^{-1}[2(540 \times 10^{-9})/(5.0 \times 10^{-6})] = 12.474° = 12°$ or $12.5°$	[1]
5(a)(iii)	peaks/maxima shown at $\theta = 0, \pm 6°$ and $\pm 12°$ (allow variance of ±1 small sq) peaks/maxima and zero intensity in between (i.e. sharp peaks & flat region in between) central peak/maxima at intensity I_0 and the other peaks all equal to or less than I_0   	[1] [1] [1]
5(b)	intensity (of all peaks / maxima) halved	[1]